The exponential curve of orderedness of cards

To what extent does orderedness of cards increase the more rounds are played in a game of Speed?

International Baccalaureate Investigation in Mathematics

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3 Words

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1 Introduction

Some patterns are hard to turn a blind eye on, especially when they might sway the outcome one way or another. Playing cards is one of my favourite ways to connect with people, and so I decided to explore the maths my favourite card game "Speed". For a while now, I've had a suspicion that the more rounds of a game we played, the more ordered the cards became, however I still wondered if this was in fact true, or if I had simply jumped to logical conclusions.

This investigation will use primary data to assess the orderedness of cards as the rounds progress, and then employ graphing and statistical strategies to analyse if my hypothesis is correct or null and void of validity.

1.1 Game explained

1.1.1 Setup

In the game of "Speed", the deck is split evenly in half, and each player sets up their half of the deck, respectively, in a layered formation, as shown below:



Figure 1: First layer - one faced up, four faced down



Figure 2: Second layer - one faced up, three faced down



Figure 3: Third layer - one faced up, two faced down



Figure 4: Fourth layer - one faced up, one faced down



Figure 5: Fifth layer - one faced up



Figure 6: Final setup of Speed for player 1



Figure 7: Final setup of Speed with both players' cards



Figure 8: Speed setup as soon as the game begins

The remainder of each half deck is placed in the centre, and now act as draw piles (as seen in Figure 7), that are played alongside onto the respective playing pile, as seen in Figure 8.

1.1.2 Rules

The aim of the game is to finish the layered cards in front of you. A card can be played if it is either one above or one below (not the same) as the card shown on the playing pile, and suits don't matter.

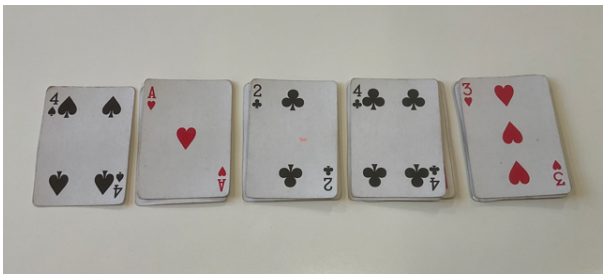


Figure 9: Good sample (more ordered) of starting cards

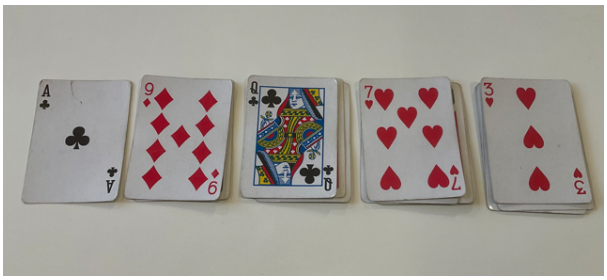


Figure 10: Bad sample (less ordered) of starting cards

Thus, the more similar in number your cards are, the more of them you can play. Each turn, both players draw a card, and play out their own cards onto either pile until neither can play anymore, then another card is drawn and the turns repeat until one player depletes their cards. At this point, either player can select the smallest pile in front of them, slap it, and claim it as part of their cards, while the loser is condemned to the larger pile. These newly claimed piles are added on top of each respective draw pile, without being shuffled, and the players set up another round within the larger game.

1.2 Definitions

- Number of rounds played (a): the rounds played within one game, with no shuffling between each one
- Cards flipped from the draw pile (b): The number of cards flipped, simultaneously, by both players

- Length of longest chain (c): the number of cards that are either one up or one down from the previous card creating a chain, applicable for this specific game

1.3 Aim and Methodology

This investigation aims to explore the relationship between the number of rounds played and the orderedness of the cards, by enacting the following steps:

1. Collecting valid and reliable data

2. Graphing the data, and assessing which regression line fits the data points best. Understanding and outlining the reasons for this pattern in the data.

3. Employing (STATS) to analyse the data

1.4 Assumptions

The findings made in this investigation are controlled and contextualised by:

1) Control variables

- All games (data) are played by the same two players
- All games were played within the span of 2 weeks to ensure minimal time for potential skill improvement in the game
- I recorded all game details myself, without any external contribution to the data

2) Ideal conditions

- Both players maintain their usual skill level throughout all games so as not to disrupt the data

2 Exploration

2.1 Initial Trials

With the aim of understanding how the number of rounds affects the orderedness of the cards in play, I first collected data on the relationship between the following variables:

Axis	Description	Variable
x	The number of rounds played	Independent
y	The cards flipped from the draw pile	Dependent

Table 1: Variables for first investigation

The cards flipped from the draw pile is a key indicator that isolates the orderedness of only the cards in play rather than assessing the orderedness of the whole deck. At this stage, I thought this would reduce the scope and therefore the complexity of the data. However, after collecting this data and graphing it on Desmos, I discovered that the relationship gave a random result that didn't indicate any kind of trend, like the one I had hypothesised.

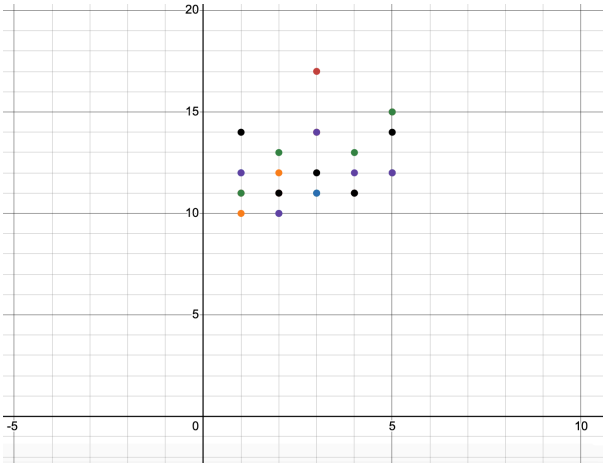


Figure 11: First Fail

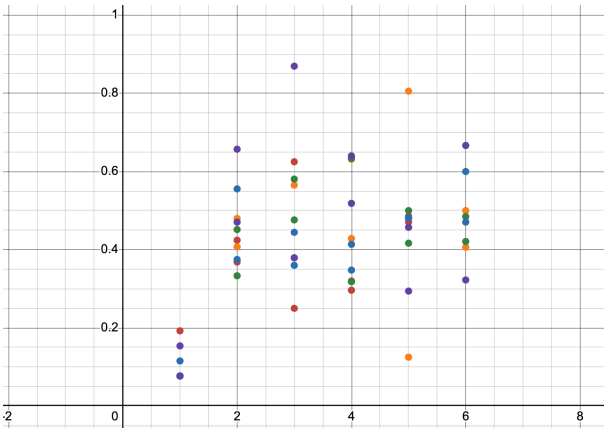


Figure 12: Second Fail

Following this setback, I tried a different approach that still assessed the number of

rounds as the independent variable, however changed the dependent variable to include the whole deck of cards. Assesssing the orderedness of the few cards faced up in front of each player could be useful for individual rounds, but the goal of assessing a whole game's progress inevitably requires analysis of the whole deck. And so, I changed the dependent variable to be the longest run that exists in each player's hand (see Table 2), to more accurately represent the orderedness of cards as it allows for cards in order, both ascending and descending, to be recorded.

In this new exploration, I used the following factors:

Axis	Description	Variable
x	The number of rounds played	Independent
y	The length of the longest chain	Dependent

Table 2: Variables for second investigation

2.2 Data Collection

I played 10 games of Speed, limiting each game to 5 rounds. I chose 10 as the number of games' data to collect, as the values for each round began to average out around that point (see Figure 13-18 below).

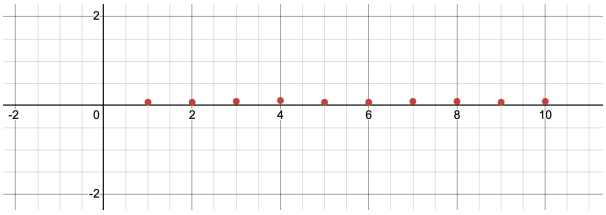


Figure 13: Round 1: graph of 10 games

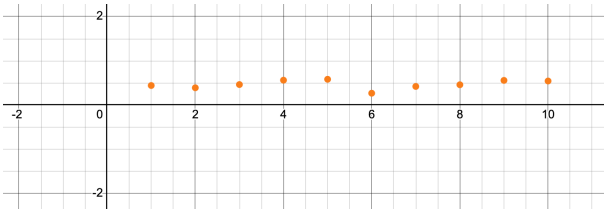


Figure 14: Round 2: graph of 10 games

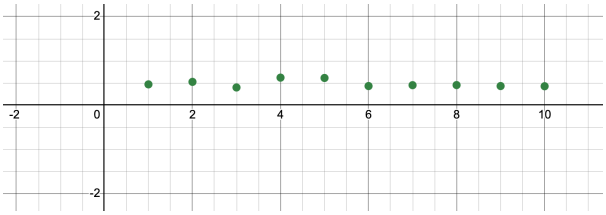


Figure 15: Round 3: graph of 10 games

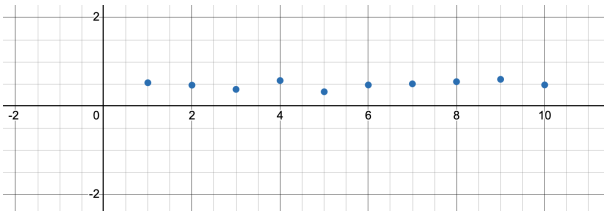


Figure 16: Round 4: graph of 10 games

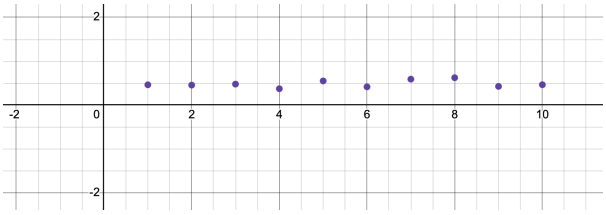


Figure 17: Round 5: graph of 10 games

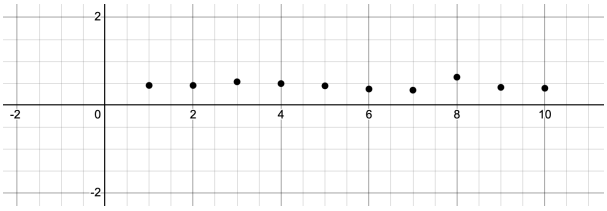


Figure 18: Round 6: graph of 10 games

I took count of the *c*-value (see Definitions) before each round, and once after the final, fifth round, producing 6 data points for each game.

Game	Round	P1 Longest Run	*P1 Total Cards*	P2 Longest Run	*P2 Total Cards*
1	1	2	*26*	2	*26*
	2	12	*25*	11	*27*
	3	11	*29*	13	*23*
	4	24	*38*	6	*14*
	5	2	*16*	29	*36*
	6	10	*20*	13	*32*
2	1	2	*26*	2	*26*
	2	14	*31*	7	*21*
	3	10	*21*	18	*31*
	4	19	*30*	7	*22*
	5	10	*24*	14	*28*
	6	8	*19*	16	*33*

Figure 19: Table of sample raw data from game 1 and 2 (see Appendix for full raw data set)

2.3 Graphing

After collecting adequate data on these variables, I found the average ratio (r), for each round from 0-5, for each player individually (r1 and r2), and then found the average between those, which presented the data set:

Game	Round	P1 Longest Run	*P1 Total Cards*	P2 Longest Run	*P2 Total Cards*	P1 Ratio (r1)	P2 Ratio (r2)	Averaged ratio (r)
1	1	2	*26*	2	*26*	0.07692307692	0.07692307692	**0.07692307692**
	2	12	*25*	11	*27*	0.48	0.4074074074	**0.4437037037**
	3	11	*29*	13	*23*	0.3793103448	0.5652173913	**0.4722638681**
	4	24	*38*	6	*14*	0.6315789474	0.4285714286	**0.530075188**
	5	2	*16*	29	*36*	0.125	0.8055555556	**0.4652777778**
	6	10	*20*	13	*32*	0.5	0.40625	**0.453125**
2	1	2	*26*	2	*26*	0.07692307692	0.07692307692	**0.07692307692**
	2	14	*31*	7	*21*	0.4516129032	0.3333333333	**0.3924731183**
	3	10	*21*	18	*31*	0.4761904762	0.5806451613	**0.5284178187**
	4	19	*30*	7	*22*	0.6333333333	0.3181818182	**0.4757575758**
	5	10	*24*	14	*28*	0.4166666667	0.5	**0.4583333333**
	6	8	*19*	16	*33*	0.4210526316	0.4848484848	**0.4529505582**

Figure 20: Table of sample data from game 1 and 2 (see Appendix for full raw set)

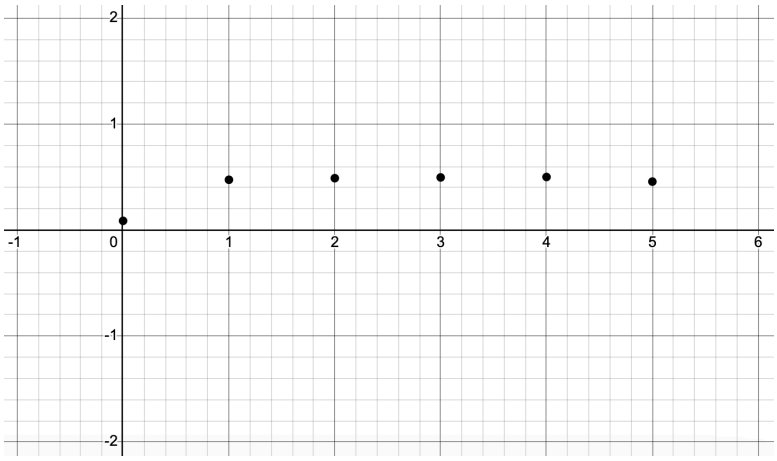


Figure 21: Graph of averaged values for each round

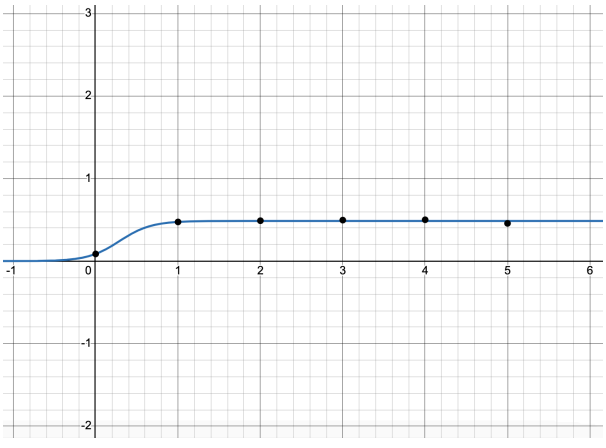


Figure 22: Graph with logistic regression line

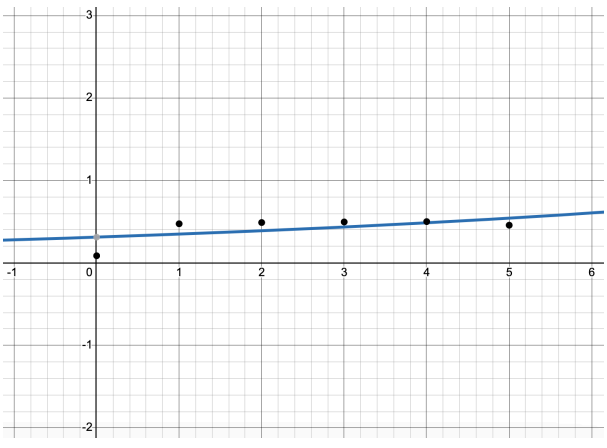


Figure 23: Graph with exponential regression line

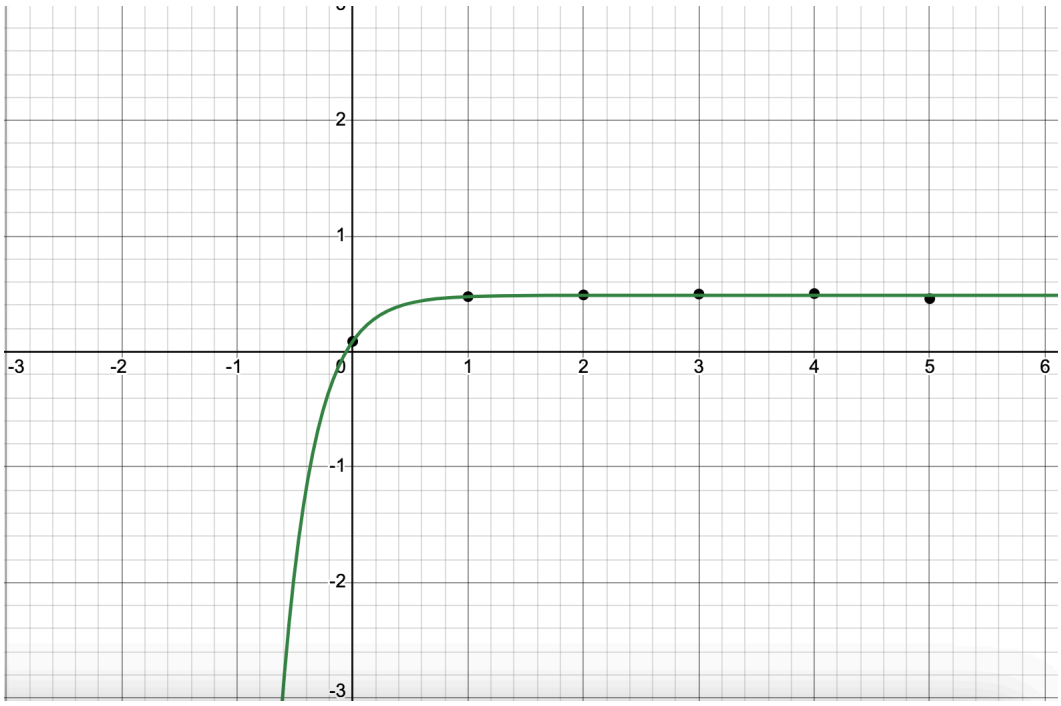


Figure 24: Graph with line of best fit

I tried various regression lines on this data (Figure 22 23), however evidently the line of best fit was the exponential graph (24) with the following equation:

$$y = a \cdot b^{cx} + d \tag{1}$$

With the following values:

a=-0.399228

b=0.150864

c=1.8832

d=0.486637

Where b and c can be grouped together, and d is expected to act as an asymptote however

The average orderliness at the start ($x = 0$), prior to any game-playing can be derived through algebra:

$$y = a \cdot b^{cx} + dy = a + dy = -0.399228 + 0.486637y = 0.087409 \tag{2}$$

The average orderliness at the end ($x = 5$), following the 5th round of game-playing can also be derived through algebra:

$$y = a \cdot b^{cx} + dy = -0.399228 \cdot 0.150864^{1.8832 \cdot 5} + 0.486637y = y = 0.087409 \tag{3}$$

This graph illustrates the extreme incline and then rounding and ultimately flat-lining of the data best, especially in comparison to the others. This curve also begins by going directly up, rather than gradually ascending, such as in Fig. 13. From this graph, we learn that the relationship between the number of rounds played and the number of cards in the longest chain is weak in the first round, however thereafter increases sharply and settles around 0.45-0.5 without dramatically increasing or decreasing, indicating n asymptotic nature of the orderedness of cards in Speed rounds. Through a real-world lens, this ultimately indicates that while the more rounds played does result in more ordered cards, this does not perpetuate the cycle entirely until the entire deck is ordered, but rather cuts off once the deck reaches a certain level of orderedness.

to do avg orderliness at the stratr using algerba avg. end asymptote orderliness using algebra what is avg extra it gets each time (round) assumption: exponential, got evidence use AI to make suggestion about maths to research

find a and d

a+d= y-int (initial orderliness) d: end-point ordelriness

b: how ordered is the pile

- look at textbook for stats